Martin **Meinel**

COMPUTATIONAL SCIENTIST · POSTDOCTORAL RESEARCHER

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Profile

Computational scientist developing AI methods for early drug target discovery and diagnostics with translational impact. Built an attention-based causal inference framework prioritizing candidate disease genes at single-cell-state resolution. Reduced 20,000 genes to a two-gene diagnostic classifier, validated across 8 international cohorts, resulting in a filed patent and a research-use-only diagnostic tool. Experienced in cross-functional collaborations spanning computational biology, clinical research, and experimental validation.

Research Experience

Cell-state-dependent eQTL mapping in single-cell genetics data

Helmholtz Munich

POSTDOCTORAL RESEARCHER — *Aug 2026 - present*

- Developing Poisson mixed-effects models to map cell-state-dependent eQTLs in population-scale single-cell genetics data (OneK1K), linking regulatory variants to disease mechanisms.

FocusMR: Causal inference for early-stage drug target identification

Helmholtz Munich & CZI Residency

PHD STUDENT — *Oct 2023 - Aug 2026*

- Developed FocusMR, an attention-based multiple-instance learning framework for Mendelian randomization that identifies candidate causal disease genes and the specific cell states in which they act, enabling more precise therapeutic targeting.
- Prioritized 34 candidate causal genes for eczema by integrating disease GWAS with population-scale single-cell genetics data (OneK1K), resolving **four loci to cell states finer than standard cell-type annotations**.
- Showed that the discovered cell states carry disease-associated biology by recovering known immune mechanisms, including PD-1 signaling, CD28 costimulation, and Th17/Th22 programs.
- Validated discovered cell states on held-out donors with Bayesian model comparison of cell-type vs. cell-state causal effects.

Gene-expression-based classifier enabling early rare cancer detection

Helmholtz Munich

PHD STUDENT — *Oct 2021 - Aug 2026*

- Developed a gene-expression-based diagnostic classifier to distinguish a rare cutaneous T cell lymphoma (early mycosis fungoides, MF) from the two most common inflammatory skin diseases (eczema and psoriasis).
- Created the biomarker discovery framework, SAFARI, revealing a **two-gene diagnostic signature** from bulk RNA-seq data comprising 20,000 protein-coding genes.
- Built a classifier with calibrated probabilities to defer clinically ambiguous cases. Achieved **91% sensitivity and 94% specificity** in the qPCR development cohort, and demonstrated generalizability across **8 independent international test cohorts**.
- Showed earlier detection than standard-of-care methods (histopathology and clinical presentation) in 4 longitudinally followed patients.
- Integrated public single-cell and spatial transcriptomic (10x Visium) datasets to identify tissue layers and cell types where biomarkers are expressed (T cells, keratinocytes).
- Filed **patent** covering the identified biomarkers and developed research-use-only diagnostic tool, Erysk, in collaboration with Dermagnostix.

Efficient implementation of deep convolutional Gaussian Processes (Master's thesis)

TU Munich

MASTER'S STUDENT — *Jan 2020 - Nov 2020*

- Implemented convolutional GPs as an uncertainty-aware alternative to CNNs (MNIST) and benchmarked imputation algorithms that compute a subset of kernel-matrix entries exactly and estimate the rest (thesis available [here]).
- Reduced training time by 40% by computing only 20% of kernel entries exactly, at a cost of 12 points of accuracy (86% vs. 98%).

Publications & Manuscripts

- **Meinel, M.**†, Langreder, N.†, et al. (2026). LCK and HOMER1 gene-expression based classifier distinguishes early mycosis fungoides from eczema and psoriasis. Manuscript completed peer review; in final editorial revision at *Science Translational Medicine*. preprint: https://www.medrxiv.org/content/10.1101/2025.10.18.25338021v1
- Garger, D.†, **Meinel, M.**†, Dietl, T., Hillig, C., Garzorz-Stark, N., Eyerich, K., de Angelis, M., Eyerich, S., Menden M.P. (2023). The impact of the cardiovascular component and somatic mutations on ageing. *Aging Cell*, https://doi.org/10.1111/acel.13957.
- Garzorz-Stark, N. †, Hillig, C.†, Maboudi Afkham, H.†, **Meinel, M.**, et al. (2026). Noncommunicable inflammatory skin diseases comprise clinically meaningful distinct endotypes as identified by nonhypothesized integration of phenotypic and transcriptome data. *Journal of Allergy and Clinical Immunology*, https://doi.org/10.1016/j.jaci.2026.08.017.
- Fischer, F., Doll, A., Uereyener, D., Roenneberg, S., Hillig, C., Weber, L., Hackert, V., **Meinel, M.**, Farnoud, A., Seiringer, P., et al. (2023). Gene expression based molecular test as diagnostic aid for the differential diagnosis of psoriasis and eczema in formalin fixed and paraffin embedded tissue, microbiopsies and tape strips. *The Journal of Investigative Dermatology*, S0022-202X(23)00156-2. https://doi.org/10.1016/j.jid.2023.02.015.
- Farnoud, A., Ohnmacht, A. J., **Meinel, M.**, & Menden, M. P. (2022). Can artificial intelligence accelerate preclinical drug discovery and precision medicine? *Expert opinion on drug discovery*, 17(7), 661–665. https://doi.org/10.1080/17460441.2022.2090540

† *symbolizes shared first authorship.*

Skills

Genomic Technologies	Single-cell RNA-seq analysis & integration (scanpy, Harmony, scVI, scANVI, scCODA) Spatial transcriptomics analysis & integration (scanpy, scANVI, 10x Visium) Statistical genetics (Mendelian randomization, GWAS, HEIDI, Colocalization, PLINK) Bulk RNA-sequencing analysis (DESeq2, edgeR, fgsea, clusterProfiler)
Machine Learning & Data Analysis	Python (NumPy, Pandas, scikit-learn, PyTorch), R, Bash
Machine Learning Methods	Attention-based multiple-instance learning, Generalized linear mixed models, Bayesian model comparisons, Uncertainty calibration
High-Performance & Cloud Computing	Hydra, Slurm, Docker, Git, AWS
Languages	English (Fluent), German (Native)

Supervision, Teaching & Peer Review

2025	Supervised three students through their master's theses and internships.	*Munich, Germany*
2025	Reviewer for the RECOMB conference.	*Seoul, South Korea*
2024	Reviewer for the Machine Learning for Health (ML4H) workshop.	*Vancouver, Canada*
2024	Teaching assistant for "Computational Modeling for System Genetics" & "Machine Learning for Biologists".	*TUM & LMU*

Honors & Awards

Patent

2025	Published patent application "Differential Diagnosis of Mycosis Fungoides: Method and System" (WO/2025/051923), using SAFARI to discover biomarkers distinguishing early MF from eczema & psoriasis.	*Munich, Germany*

Chan Zuckerberg Initiative Residency

2024	Selected for CZI residency for developing cell-context-aware drug target discovery methods to identify cellular contexts of causal genes for precise therapeutic targeting.	*Redwood City, USA*

Oral presentations

2025	Oral presentation at NeurIPS AI4D3 workshop on how to identify cellular contexts of causal genes enhancing early-stage drug target discovery.	*San Diego, USA*
2025	Oral presentation at EORTC CLTG about the identification and clinical translation of biomarkers in early MF.	*Athens, Greece*
2023	Invited speaker session about "Biocomputational Analysis of Big Data" at the EADV-ESDR Summer Research Workshop: Big Data and Translation in Dermatology.	*Freiburg, Germany*

Poster presentations

2025	Highlight poster with oral presentation on cell-state-aware single-cell Mendelian randomization at RECOMB Genetics.	*Seoul, South Korea*
2023	Presented the SAFARI framework for rare disease biomarker discovery at the ML4H workshop.	*New Orleans, USA*

Education

PhD in Computational Biology & Statistical Genetics — *Munich, Germany*
Helmholtz Center Munich | LMU Munich | Chan Zuckerberg Initiative Residency — *Oct 2021 - Aug 2026*
- Supervised by Dr. Michael Menden, Dr. Natalie Garzorz-Stark & Dr. Francesco Paolo Casale

M.Sc. in Data Engineering & Analytics — *Munich, Germany*
Technical University of Munich — *Oct 2018 - Nov 2020*

B.Sc. in Applied Computer Science — *Bayreuth, Germany*
University of Bayreuth — *Oct 2014 - Oct 2017*

Working experience

Opitz Consulting GmbH — *Munich, Germany*
Software Engineer Working student — *Oct 2019 - Oct 2020*
- Built a Docker skeleton for Apache Zeppelin, enabling users to start a docker container running Apache Zeppelin on OpenShift with one click.
- Created a web application for Allianz IDS GmbH, allowing employees to track, assign, and assess investment orders within the department.